Handling Missing Data in Machine Learning: A Comparison of Imputation Strategies and Missingness Patterns

1. Introduction

Missing values are common in real datasets, especially clinical ones, where tests are skipped, records are incomplete or measurements fail. Most machine learning models cannot use incomplete rows directly, so missing values are usually filled in (imputed) first. The choice of imputation method can change model accuracy, and how the data went missing matters too. Rubin (1976) distinguished data that is missing completely at random (MCAR), missing depending on other observed variables (MAR), and missing depending on the unobserved value itself (MNAR).

This project asks two questions:

1. How much does missing data reduce the accuracy of a classifier, and does the imputation method change this?
2. Does the missingness pattern (MCAR vs MNAR) change the result?

2. Data and method

Data. The Breast Cancer Wisconsin (Diagnostic) dataset from scikit-learn has 569 patients. Six measurements were used (mean radius, texture, perimeter, area, smoothness and compactness) to classify tumours as malignant or benign.

Simulating missing data. The complete dataset was treated as the truth. Between 10% and 50% of values were deleted, either at random (MCAR) or with larger values more likely to be deleted (MNAR).

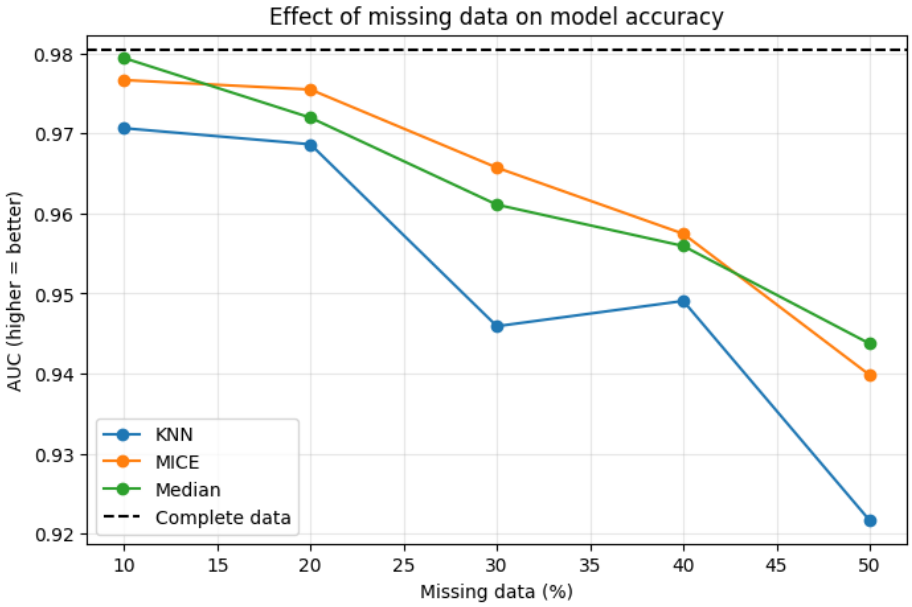
Imputation methods. Three methods were compared: median imputation, K-nearest-neighbours (KNN, k = 5) and MICE-style iterative imputation (scikit-learn's IterativeImputer, after van Buuren and Groothuis-Oudshoorn, 2011).

Model and evaluation. A Random Forest (100 trees) was used. Performance was measured with AUC (1.0 is perfect) using 5-fold cross-validation, with the imputer inside a scikit-learn Pipeline so it learned only from training folds and no data leaked. A second analysis used an 80/20 train/test split at 30% missing, shown with confusion matrices and ROC curves.

3. Results

Effect of missing data. The complete data model reached an AUC of 0.981. As missingness increased, AUC fell for every method, to between 0.922 and 0.944 at 50% missing.

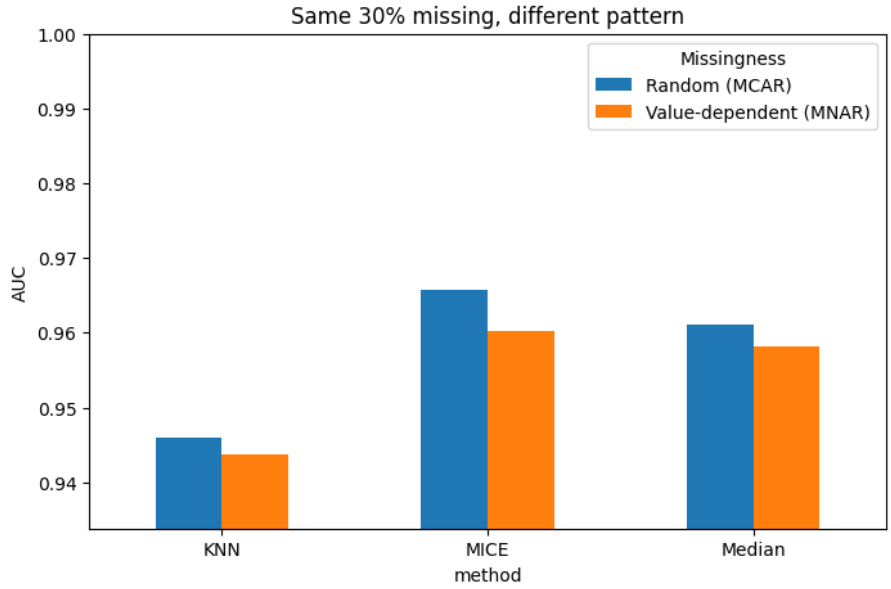
Missing %	Median	KNN	MICE
10	0.979	0.971	0.977
20	0.972	0.969	0.975
30	0.961	0.946	0.966
40	0.956	0.949	0.957
50	0.944	0.922	0.940
Average	0.962	0.951	0.963



Comparing methods. KNN had the lowest AUC at every missing level, with an average of 0.951. Median (0.962) and MICE (0.963) were almost identical. At 50% missing, Median was slightly higher (0.944 vs 0.940), while MICE was slightly higher at 20%, 30% and 40%.

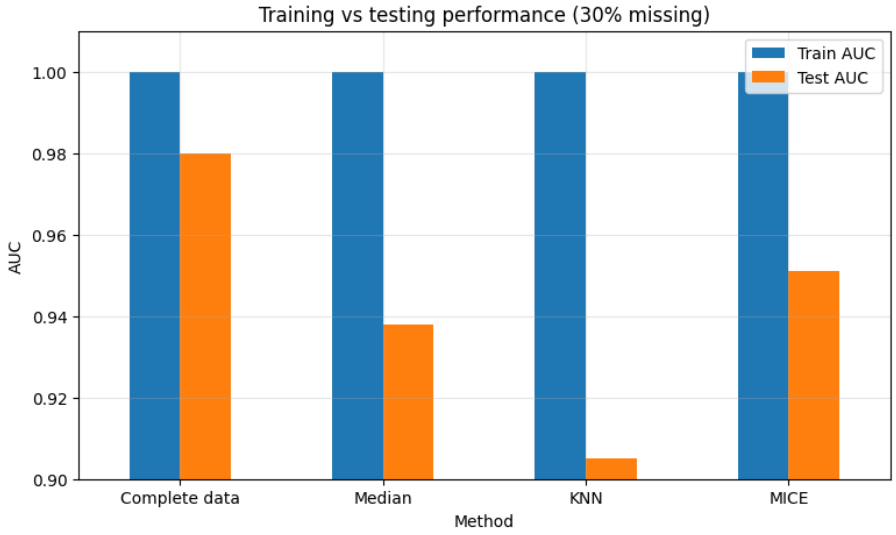
Missingness pattern. At 30% missing, MNAR gave a slightly lower AUC than MCAR for MICE (0.960 vs 0.966) and Median (0.958 vs 0.961), and nearly the same for KNN (0.944 vs 0.946). The differences were at most 0.006.

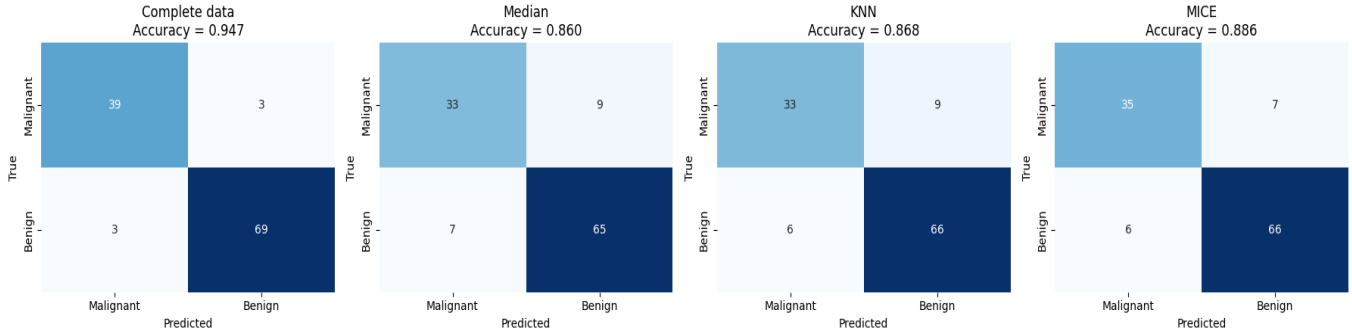
Method	Random (MCAR)	Value-dependent (MNAR)
Median	0.961	0.958
KNN	0.946	0.944
MICE	0.966	0.960

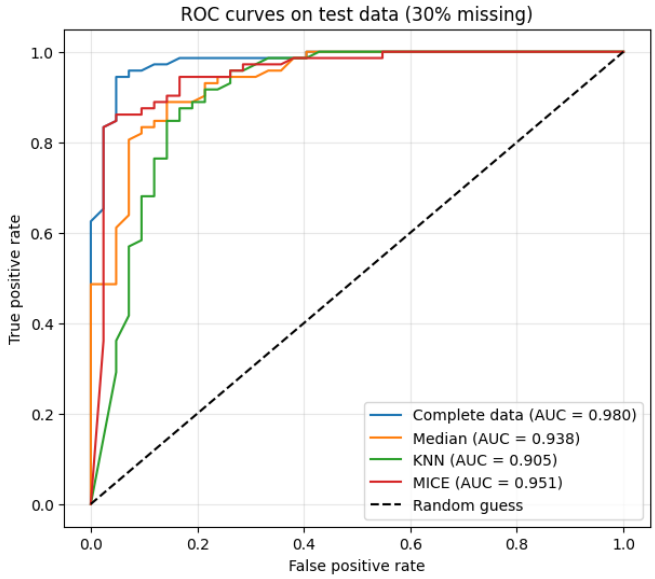


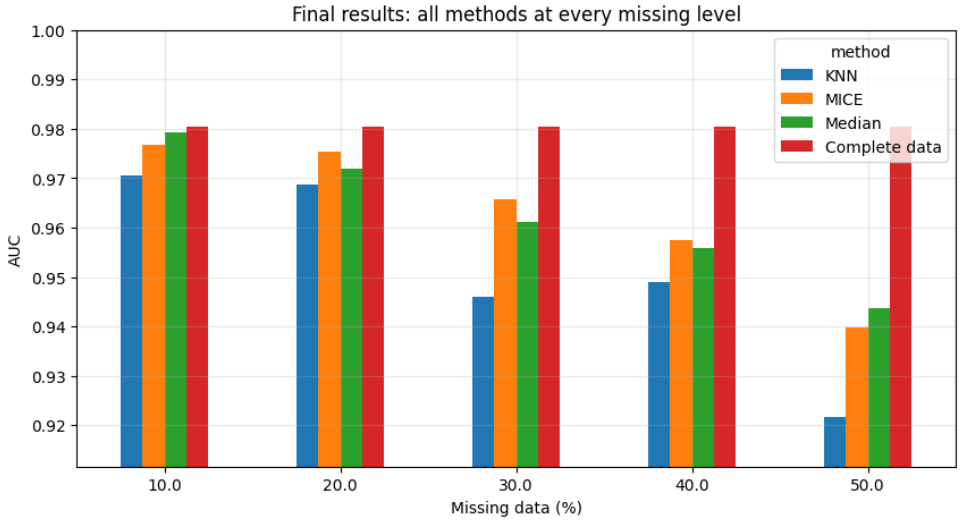
Training vs testing. On the 80/20 split with 30% missing, every model had a training AUC of 1.000. Test AUC was 0.980 for complete data, 0.951 for MICE, 0.938 for Median and 0.905 for KNN.

Model	Train AUC	Test AUC
Complete data	1.000	0.980
Median	1.000	0.938
KNN	1.000	0.905
MICE	1.000	0.951









4. Discussion

The main result is that missing data lowers accuracy, and the loss grows as more values go missing. Even so, the model held up well. At 50% missing, AUC was still above 0.92 for all methods. This dataset has strongly correlated measurements (radius, perimeter and area describe similar tumour size), so the information in deleted values is partly recoverable from the remaining columns. That may explain why simple median filling performed as well as MICE.

KNN's lower scores may be because it relies on distances between patients, which become less reliable when many values are missing. This is a possible explanation and was not tested here.

The Random Forest scored 1.000 on training data for all models, which reflects its ability to fit the training set closely. The test AUC is the honest measure, and it shows how much accuracy was lost on unseen patients.

The small gap between MCAR and MNAR suggests that, for this dataset and the 30% level, the pattern mattered less than the amount of missing data. Given the limitations below, this should not be generalised.

5. Limitations and future work

- Only one small dataset, six features and one model were used.
- Missing values were simulated, so real clinical missingness may behave differently.
- Each missing level used a single random mask with no repeated runs, so small differences such as Median vs MICE (about 0.005 AUC) should not be treated as meaningful. This also explains why the AUC does not always fall smoothly (for example, KNN is 0.946 at 30% and 0.949 at 40%).
- The train/test analysis used one split with about 114 test patients, so those scores are noisy.

Future work: repeat with multiple random seeds and report variability, test more datasets and models, add calibration and explainability analysis, and extend the idea to missing information in medical.

References

- Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63(3), 581 to 592.
- van Buuren, S., & Groothuis-Oudshoorn, K. (2011). mice: Multivariate imputation by chained equations in R. *Journal of Statistical Software*, 45(3).
- Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825 to 2830.